

Quality Control System Implementation

Overview

A comprehensive quality control system for chicken processing that tracks quality checks, corrective actions, compliance, and generates certificates for batches.

Requirements Met

1. Record quality check results (pass/fail) for batches

- ☒ Quality check model with parameters, measurements, and results
- ☒ Integration with batch system
- ☒ Pass/fail/conditional result tracking

2. Document corrective actions for failed checks

- ☒ Corrective action model with assignment, due dates, and status tracking
- ☒ Severity levels (critical, major, minor, observation)
- ☒ Evidence and verification tracking

3. Track compliance with food safety standards

- ☒ Compliance type integration
- ☒ Regulatory references and inspection tracking
- ☒ Audit trail for all quality operations

4. Generate quality certificates for batches

- ☒ Certificate generation for passed checks
- ☒ Certificate numbering and validation periods
- ☒ QR code and document URL support

5. Maintain audit trails for regulatory inspections

- ☒ Comprehensive audit trail on all quality checks
- ☒ Action tracking with timestamps and user attribution
- ☒ Change history for traceability

6. Schedule regular quality checks based on production schedule

- ☒ Quality check scheduling model
- ☒ Frequency-based scheduling (daily, weekly, monthly, etc.)
- ☒ Production stage-based scheduling

Architecture

Models

1. **QualityCheck** - Core quality check entity

- Batch association
- Check types (visual inspection, microbiological test, etc.)
- Parameter measurements and results
- Audit trail
- Corrective actions

2. **QualityCheckSchedule** - Scheduled quality checks

- Frequency configuration
- Applicable batch types
- Parameter standards

3. **QualityCertificate** - Generated certificates

- Certificate details and numbering
- Test results summary
- Validity periods

4. **QualityDashboard** - Analytics and reporting

- Pass/fail rates
- Severity distribution
- Compliance status

Services

- **QualityControlService** - Core business logic
 - Create and update quality checks
 - Submit results and calculate outcomes
 - Manage corrective actions
 - Generate certificates
 - Search and filtering

Routes

- [/api/quality/checks](#) - CRUD operations for quality checks
- [/api/quality/checks/:id/results](#) - Submit check results
- [/api/quality/checks/:id/corrective-actions](#) - Manage corrective actions
- [/api/quality/batches/:batchId/checks](#) - Get checks for a batch
- [/api/quality/check-types](#) - Get available check types
- [/api/quality/check-statuses](#) - Get check statuses
- [/api/quality/severity-levels](#) - Get severity levels
- [/api/quality/corrective-action-statuses](#) - Get corrective action statuses
- [/api/quality/validate/*](#) - Validation endpoints
- [/api/quality/health](#) - Health check

Key Features

1. Parameter Evaluation

- Automatic evaluation of measured values against standards
- Tolerance-based pass/fail/warning determination
- Overall result calculation from individual parameters

2. Corrective Action Management

- Assignment to specific users
- Due date tracking
- Status workflow (open → in progress → completed → verified → closed)
- Evidence attachment

3. Audit Trail

- Every action logged with timestamp and user
- Change tracking for modifications
- Regulatory compliance ready

4. Batch Integration

- Quality checks linked to batches
- Automatic batch status updates (e.g., quarantined for failed checks)
- Quality check history per batch

5. Search and Filtering

- Filter by batch, check type, status, date range
- Pagination support
- Sorting options

Data Model

Quality Check Types

- **VISUAL_INSPECTION** - Visual quality assessment
- **MICROBIOLOGICAL_TEST** - Microbial contamination testing
- **CHEMICAL_ANALYSIS** - Chemical composition testing
- **PHYSICAL_TEST** - Physical property testing
- **SENSORY_EVALUATION** - Sensory quality assessment
- **PACKAGING_INTEGRITY** - Packaging quality testing
- **TEMPERATURE_CHECK** - Temperature monitoring
- **WEIGHT_VERIFICATION** - Weight accuracy checking
- **LABELING_ACCURACY** - Label information verification

Quality Check Statuses

- **PASSED** - All parameters within acceptable range
- **FAILED** - One or more parameters outside acceptable range

- **PENDING** - Check created but not performed
- **IN_PROGRESS** - Check being performed
- **CANCELLED** - Check cancelled

Severity Levels

- **CRITICAL** - Immediate safety concern
- **MAJOR** - Significant quality issue
- **MINOR** - Minor quality deviation
- **OBSERVATION** - Observation without immediate impact

Corrective Action Statuses

- **OPEN** - Action created, not started
- **IN_PROGRESS** - Action being implemented
- **COMPLETED** - Action implemented
- **VERIFIED** - Action verified as effective
- **CLOSED** - Action closed

Testing

Run the quality control tests:

```
cd chicken-processing-backend  
node test-quality.js
```

Expected output:

- Quality check types endpoint
- Quality check statuses endpoint
- Severity levels endpoint
- Corrective action statuses endpoint
- Search functionality
- Validation endpoints
- Health check

Integration Points

1. Batch System

- Quality checks linked to batches via **batchId**
- Batch status updates based on quality results
- Quality check history in batch records

2. User System

- User authentication for all operations

- User attribution in audit trails
- Assignment of corrective actions to users

3. Compliance System

- Compliance type integration
- Regulatory standard references
- Certificate generation

4. Notification System (Future)

- Alerts for critical failures
- Reminders for overdue checks
- Notifications for corrective action assignments

Future Enhancements

1. **Automated Scheduling** - Automatic creation of scheduled checks
2. **Notification System** - Email/SMS alerts for quality issues
3. **Reporting Dashboard** - Visual analytics and trend analysis
4. **Mobile App** - Field quality checks via mobile devices
5. **Integration with IoT** - Real-time sensor data integration
6. **Supplier Quality Tracking** - Track quality by supplier
7. **Predictive Analytics** - Predict quality issues before they occur

Files Created

1. `src/models/QualityControl.ts` - Data models and interfaces
2. `src/services/qualityControlService.ts` - Business logic
3. `src/routes/quality.ts` - API routes
4. `test-quality.js` - Test script
5. Updated `src/app.ts` - Route registration

Usage Example

```
// Create a quality check
const checkData = {
  batchId: 'batch-123',
  checkType: 'VISUAL_INSPECTION',
  scheduledDate: new Date(),
  parameters: [
    {
      name: 'color',
      unit: 'rating',
      standardValue: 5,
      measuredValue: 4,
      tolerance: '1'
    }
  ]
}
```

```

    ],
    performedBy: 'user-123',
    lastUpdatedBy: 'user-123'
  };

  // Submit results
  const results = {
    parameters: [
      {
        name: 'color',
        measuredValue: 4,
        result: 'pass',
        notes: 'Slightly pale but acceptable'
      }
    ],
    overallResult: 'pass',
    notes: 'Visual inspection passed'
  };

  // Create corrective action for failed check
  const actionData = {
    qualityCheckId: 'check-123',
    description: 'Adjust lighting for better color assessment',
    assignedTo: 'user-456',
    dueDate: new Date('2024-12-31'),
    priority: 'MINOR',
    lastUpdatedBy: 'user-123'
  };

```

Compliance Benefits

1. **Traceability** - Complete audit trail for regulatory inspections
2. **Documentation** - Comprehensive records of all quality checks
3. **Corrective Action Tracking** - Systematic approach to addressing issues
4. **Certificate Generation** - Official documentation for compliant batches
5. **Real-time Monitoring** - Immediate identification of quality issues

This implementation provides a robust foundation for food safety and quality management in chicken processing operations.